



Cough in Adults

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Cough can be a protective reflex that prevents aspiration and expels particles and irritants from the airway. However, it can become uncontrolled or disabling and is one of the most common symptoms prompting physician visits. (See also [Cough in Children](#).)

Likely causes of cough (see table [Some Causes of Cough](#)) differ depending on whether the symptom is acute (present for less than 3 weeks), subacute (present for 3 to 8 weeks), or chronic (present for more than 8 weeks).

In **acute cough**, the most common causes are:

- Upper respiratory infection ([URI](#)), including acute bronchitis
- Acute rhinosinusitis
- [Allergic rhinitis](#)
- Acute exacerbation of [chronic obstructive pulmonary disorder](#) (COPD)
- [Pneumonia](#)
- [Pertussis](#)
- Acute exacerbation of [heart failure](#)

In **subacute cough**, the most common causes is:

- Airway hyperresponsiveness after resolution of a viral or bacterial respiratory infection (ie, postinfection cough)

In **chronic cough**, the most common causes are:

- Postnasal drip or upper airway cough syndrome (of allergic, nonallergic, or infectious origin)
- [Gastroesophageal reflux](#)
- [Asthma](#)
- [Chronic bronchitis](#)
- [Chronic obstructive pulmonary disease](#) (COPD)
- Angiotensin-converting enzyme (ACE) inhibitors and, less often, angiotensin II receptor blockers
- Chronic refractory and unexplained chronic cough

The causes of [cough in children](#) are similar to those in adults, but [asthma](#) may be more common in children. Foreign body aspiration, uncommon in adults except in those with impaired development, dementia, or swallowing dysfunction, is more common in children.

Other rarer causes of cough may be present. Purely psychogenic cough is a diagnosis of exclusion. However, patients with chronic cough may develop a secondary reflex or psychogenic component to their cough. Protracted coughing may injure the bronchial mucosa, which may trigger more coughing. Unexplained chronic cough refers to cough that persists in patients in whom a comprehensive investigation has not revealed a likely etiology (1). Refractory chronic cough refers to a persistent cough after treatment for cough-associated conditions.

TABLE		
Some Causes of Cough		
Cause	Suggestive Findings	Diagnostic Approach
Acute Cough		
Foreign body*	Sudden onset in a young child who has no URI or constitutional symptoms	Chest radiograph (inspiratory and expiratory views) Bronchoscopy
Heart failure *	Dyspnea Fine inspiratory crackles Extrasystolic heart sound Elevated jugular venous pressure Dependent peripheral edema Orthopnea Paroxysmal nocturnal dyspnea	Chest radiograph Brain (B-type) natriuretic peptide level Transthoracic echocardiography
Pneumonia (viral, bacterial, aspiration, rarely fungal)	Fever Productive cough Dyspnea Pleuritic chest pain Localized bronchial breath sounds or egophony	Chest radiograph Cultures (eg, sputum, pleural fluid, blood, possibly bronchial washings) in seriously ill patients and patients with hospital-acquired pneumonia

Cause	Suggestive Findings	Diagnostic Approach
	Inspiratory crackles Hypoxia on pulse oximetry	Multiplex PCR panel
Postnasal drip (allergic or infectious origin)‡	Headache Sore throat Nausea Cobblestoning of posterior oropharynx Pale, boggy, swollen nasal mucosa Frequent clearing of the throat	History and physical examination Response to empiric antihistamine, decongestant, or nasal ipratropium therapy CT of the sinuses if diagnosis is unclear
Pulmonary embolism *	Pleuritic chest pain Dyspnea Tachycardia	CT angiography Less often, ventilation/perfusion scanning and possibly pulmonary arteriography
URI , including acute bronchitis †	Rhinorrhea Red, swollen nasal mucosa Sore throat Malaise	History and physical examination
Chronic Cough		
ACE inhibitors	Dry, persistent nocturnal cough that may occur within days or months after initiation of ACE inhibitor therapy	Gauging clinical response to stopping ACE inhibitor
Aspiration	Cough after eating or drinking Wheezing	Chest radiograph Sometimes modified barium pharyngography Bronchoscopy
Asthma (including cough variant)	Cough in response to various precipitating factors (eg, allergens, cold	Pulmonary function testing

Cause	Suggestive Findings	Diagnostic Approach
	air, exercise, viral infection) Possibly wheezing and dyspnea	Methacholine challenge Response to empiric bronchodilator therapy
<u>Chronic bronchitis</u> (a subset of COPD)‡	Productive cough on most days of the month or for 3 months of the year for 2 successive years in a patient with known COPD or smoking history Frequent clearing of the throat Dyspnea	Chest radiograph Pulmonary function testing
<u>COPD</u> (chronic obstructive pulmonary disease)‡	Known diagnosis of COPD Decreased breath sounds Wheezing Dyspnea Pursed lip breathing Use of accessory respiratory muscles Tripod positioning of the arms against the legs or examination table	History and physical examination Pulmonary function testing Imaging of the chest (radiograph or CT)
<u>Gastroesophageal reflux</u>	Burning chest or abdominal pain that tends to worsen with consumption of certain foods, certain activities, or certain positions Sour taste, particularly on awakening Halitosis Hoarseness Chronic nocturnal or early	Response to empiric H2 blocker or proton pump inhibitor therapy Sometimes esophageal manometry or pH probe

Cause	Suggestive Findings	Diagnostic Approach
	morning cough	
Hyperresponsive airways after resolution of respiratory tract infection‡	Dry, nonproductive cough that may persist for weeks or months after an acute respiratory tract infection	Typically chest radiograph
Interstitial lung disease	Dyspnea of gradual onset Dry cough Inspiratory crackles History of drug (medication or illicit)§ or occupational exposure	Chest radiograph High-resolution CT Pulmonary function testing
<u>Pertussis</u>	Repeated bouts of ≥ 5 rapidly consecutive, forceful coughs during a single expiration, followed by a hurried and deep inspiration (whoop) or posttussive emesis	Cultures of nasopharyngeal specimens
Rhinosinusitis‡	Headache Sore throat Cobblestoning of posterior oropharynx Pale, boggy, swollen nasal mucosa	History and physical examination Response to empiric antihistamine or decongestant therapy Sometimes allergy testing
Tuberculosis (<u>TB</u>) Fungal infections	Atypical symptoms (eg, weight loss, fever, hemoptysis, night sweats) Exposure history Immunocompromise (eg, use of TNF inhibitors)	Chest radiograph Tuberculin skin testing or interferon gamma release assay; if positive, sputum cultures and stains for acid-fast bacilli and fungi Sometimes chest CT or bronchoalveolar lavage
<u>Tumor</u> *	Atypical symptoms (eg, weight loss, low-grade	Chest radiograph If positive, chest CT and bronchoscopic biopsy

Cause	Suggestive Findings	Diagnostic Approach
	fever, hemoptysis, night sweats) Change in chronic cough Lymphadenopathy	
* Indicates rare causes of cough. † Streptococcal pharyngeal infections notably may not cause cough. ‡ Also a cause of subacute cough. § See table Substances With Toxic Pulmonary Effects. ACE = angiotensin-converting enzyme; CT = computed tomography; PCR = polymerase chain reaction; TB = tuberculosis; TNF = tumor necrosis-factor; URI = upper respiratory infection.		

General reference

1. [Irwin RS, Madison JM](#): Unexplained or Refractory Chronic Cough in Adults. *N Engl J Med* 392(12):1203–1214, 2025. doi:10.1056/NEJMra2309906

Pathophysiology of Cough

Cough is often a normal physiologic response to certain triggers (eg, inhaled pollutants, cold and dry air, allergens, fine particulate matter). The neural pathways activated in cough form a complex reflex arc encompassing afferent (sensory), central (brain), and efferent (motor) pathways (1). A multidirectional feedback loop, mediated through the brainstem respiratory network, also called the cough center, likely influences the afferent and efferent arms. Coughing may be voluntary or involuntary.

Chronic cough may be due to the activation of myriad receptors in the body (2). Some key receptors implicated are P2X3 purinergic receptors, transient receptor potential vanilloid 1 (TRPV1), transient receptor potential ankyrin 1 (TRPA1), acid-sensing ion channels (ASICs), bradykinin B2 receptors, prostaglandin receptors, and neurokinin-1 (NK1) receptors.

Pathophysiology references

1. [Canning BJ, Chang AB, Bolser DC, et al](#): Anatomy and neurophysiology of cough: CHEST Guideline and Expert Panel report. *Chest* 146(6):1633–1648, 2014. doi:10.1378/chest.14-1481

2. [Smith JA, Woodcock A](#). Chronic Cough. *N Engl J Med* 2016;375(16):1544-1551.
doi:10.1056/NEJMcp1414215

Evaluation of Cough

History

History of present illness should cover the onset, duration, and characteristics of the cough (eg, whether dry or productive of sputum or blood, and whether it is accompanied by [dyspnea](#), [chest pain](#), or both). Asking about precipitating factors (eg, cold air, strong odors) and the timing of the cough can be revealing (eg, cough occurring primarily when recumbent may suggest [gastroesophageal reflux](#) or [heart failure](#) as a cause).

Review of systems should seek symptoms stemming from possible causes, including rhinorrhea, postnasal drip, and sore throat (upper respiratory infection [[URI](#)]); fever, chills, and pleuritic chest pain ([pneumonia](#)); night sweats and weight loss ([tumor](#), tuberculosis [[TB](#)]); heartburn ([gastroesophageal reflux](#)); and dysphagia or choking episodes while eating or drinking (aspiration).

Past medical history should note recent respiratory infections (ie, within previous 1 to 2 months); history of [allergies](#), [asthma](#), chronic obstructive pulmonary disease ([COPD](#)), and [gastroesophageal reflux disease](#); risk factors for (or known) [TB](#) or [HIV infection](#); and smoking history. Medication use history should specifically include use of angiotensin-converting enzyme (ACE) inhibitors. Patients with chronic cough should be asked about exposure to potential respiratory irritants or allergens, occupational exposures, and travel to or residence in regions with endemic fungal illness. Patients should also be asked if the cough is seasonal or perennial.

Physical examination

Vital signs should be reviewed for the presence of tachypnea, hypoxia, and fever.

General examination should look for signs of respiratory distress and chronic illness (eg, wasting, lethargy).

Examination of the nose and throat should focus on appearance of the nasal mucosa (eg, color, congestion) and the presence of discharge (external or in posterior pharynx). Ears should be examined for foreign bodies, masses, or signs of infection.

The cervical and supraclavicular areas should be inspected and palpated for lymphadenopathy.

A full lung examination should be performed, particularly including adequacy of air entry and exit; symmetry of breath sounds; and the presence of crackles, wheezes (diffuse or localized), or both. Signs of consolidation (eg, decreased breath sounds, egophony, dullness to percussion) should be sought.

Red flags

The following findings are of particular concern:

- Dyspnea
- Hemoptysis
- Weight loss
- Persistent fever
- Risk factors for TB or HIV infection

Interpretation of findings

Some findings point to particular diagnoses (see table [Some Causes of Cough](#)) (1).

Other important findings are less specific. For example, the color (eg, yellow, green) and thickness of sputum do not help differentiate bacterial from other causes. [Wheezing](#) may occur with several causes (eg, asthma, or heart failure).

[Hemoptysis](#) in small amounts may occur with severe cough of many etiologies, although larger amounts of hemoptysis suggest [bronchitis](#), [bronchiectasis](#), [TB](#), or primary [lung cancer](#).

Fever, night sweats, and weight loss may occur with many chronic infections as well as with cancer.

Testing

Pulse oximetry and chest radiograph should be performed in patients with red flag findings of dyspnea or hemoptysis and patients in whom suspicion of pneumonia or other parenchymal lung disease is high. A chest radiograph and testing for TB and HIV infection should be performed in patients with weight loss or risk factors for those infections.

For many patients without red flag findings, clinicians can base the diagnosis on history and physical examination findings and begin treatment without testing. For patients without a clear cause but no red flag findings, many clinicians empirically begin treatment for postnasal drip (eg, antihistamines, nasal glucocorticoid sprays, or nasal anticholinergic sprays), airway hyperreactivity (eg, short-acting beta-adrenergic agonist and/or inhaled corticosteroid [glucocorticoid], or gastroesophageal reflux disease [eg, proton pump inhibitors, H2 blockers]). An adequate clinical response to these interventions usually precludes the need for further evaluation.

Chest radiography should be performed in patients with chronic cough in whom presumptive treatment is ineffective. If the radiograph findings are unremarkable, many clinicians sequentially test for asthma (pulmonary function tests with methacholine challenge if standard spirometry is normal), sinus disease (sinus CT), and gastroesophageal reflux disease (esophageal pH monitoring).

Sputum culture is helpful for patients with a possible indolent infection, such as pertussis, TB, or nontuberculous mycobacterial infection.

Chest CT and possibly bronchoscopy should be performed in patients in whom lung cancer or another bronchial tumor is suspected (eg, patients with a long smoking history, nonspecific constitutional signs) and in patients in whom empiric therapy has failed and who have inconclusive findings on preliminary testing.

Evaluation reference

1. [Morice AH, Millqvist E, Bieksiene K, et al](#): ERS guidelines on the diagnosis and treatment of chronic cough in adults and children. *Eur Respir J* 55(1): 1901136, 2020.
doi: 10.1183/13993003.01136-2019

Treatment of Cough

Treatment is management of the cause (1).

There is little evidence to support the use of cough suppressants or mucolytic agents. Coughing is an important mechanism for clearing secretions from the airways and can assist in recovery from respiratory infections. Therefore, caution is advised when prescribing cough suppressants, reserving them primarily for patients with upper respiratory infections or those whose cough persists despite other treatment for the underlying disorder. Cough suppressants may help some patients with chronic cough who have a reflex or psychogenic component to their cough or who develop bronchial mucosal injury.

Antitussives depress the medullary cough center ([dextromethorphan](#) and [codeine](#)) or anesthetize stretch receptors of vagal afferent fibers in bronchi and alveoli ([benzonatate](#)). Dextromethorphan, a congener of the opioid levorphanol, is effective as a tablet or syrup. Codeine has antitussive, analgesic, and sedative effects, but dependence is a potential problem, and nausea, vomiting, constipation, and tolerance are common adverse effects. Other opioids ([hydrocodone](#), [hydromorphone](#), [methadone](#), [morphine](#)) have antitussive properties but are avoided because of high potential for dependence and abuse. Benzonatate, a congener of tetracaine is available in liquid-filled capsules.

Expectorants are thought to decrease viscosity and facilitate expectoration (coughing up) of secretions but are of limited, if any, benefit in most circumstances. [Guaifenesin](#) is most commonly used because it has no serious adverse effects, but multiple expectorants exist, including bromhexine, and saturated solution of [potassium iodide](#) (SSKI). Aerosolized expectorants such as [acetylcysteine](#), DNase ([dornase alfa](#)), and hypertonic saline are generally reserved for hospital-based treatment of cough in patients with [bronchiectasis](#) or [cystic fibrosis](#). Ensuring adequate hydration may facilitate expectoration, as may inhalation of steam, although randomized trials of their efficacy are lacking.

Topical treatments, such as acacia, licorice, [glycerin](#), honey, and wild cherry cough drops or syrups (demulcents), are locally and perhaps emotionally soothing, but their use is not supported by scientific evidence.

Protussives, which stimulate cough, are indicated for such disorders as cystic fibrosis and bronchiectasis, in which a productive cough is thought to be important for airway clearance and preservation of pulmonary function. DNase (eg, dornase alpha) or hypertonic saline is given in conjunction with chest physical therapy and postural drainage to promote cough and expectoration. This approach is beneficial in cystic fibrosis but not in most other causes of chronic cough.

Bronchodilators, such as [albuterol](#) and [ipratropium](#) or inhaled corticosteroids (glucocorticoids), can be effective for cough after URI and in cough-variant asthma.

Neuromodulators (eg, [gabapentin](#), [pregabalin](#)) are considered treatment options for refractory chronic cough ([2](#), [3](#)).

Multiple novel agents (eg, P2X3 receptor antagonists, neurokinin-1 antagonists) to treat refractory chronic cough are being studied as the neurobiology of the cough reflex is better characterized ([4](#)).

Treatment references

1. [Morice AH, Millqvist E, Bieksiene K, et al](#): ERS guidelines on the diagnosis and treatment of chronic cough in adults and children. *Eur Respir J* 55(1):1901136, 2020. doi: 10.1183/13993003.01136-2019
2. [Ryan NM, Birring SS, Gibson PG](#): Gabapentin for refractory chronic cough: a randomised, double-blind, placebo-controlled trial. *Lancet* 380(9853):1583–1589, 2012. doi:10.1016/S0140-6736(12)60776-4
3. [Vertigan AE, Kapela SL, Ryan NM, Birring SS, McElduff P, Gibson PG](#): Pregabalin and Speech Pathology Combination Therapy for Refractory Chronic Cough: A Randomized Controlled Trial. *Chest* 149(3):639–648, 2016. doi:10.1378/chest.15-1271
4. [Smith JA](#): The Therapeutic Landscape in Chronic Cough. *Lung* 202(1):5–16, 2024. doi:10.1007/s00408-023-00666-y

Key Points

- Danger signs include respiratory distress, chronic fever, weight loss, and hemoptysis.
- Clinical diagnosis is usually adequate.
- Occult gastroesophageal reflux disease should be remembered as a possible cause.
- Antitussives and expectorants should be used selectively; neuromodulators may be added in chronic cases of refractory cough.

Drug Information for the Topic



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